

Two Ways a Design Fails: When Does a Synthetic LLM Panel See the Danger?

A Backend-Sensitivity Audit of Synthetic-User Interface-Risk Measurement

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Abstract

An AI-advice interface can fail in (at least) two ways: it can make human reliance *heterogeneous and user-sensitive* (an *over-dispersion* failure), or make users *over-rely on a confidently wrong AI* under a coercive framing (a *dark-pattern* failure). Detecting either normally needs an expensive human study, so a tempting shortcut is to screen the interface first with a multi-agent LLM panel that plays a synthetic cohort of users. Most debate about such panels asks whether they *match* humans. We ask a question that comes first and is rarely tested: is the panel’s risk *measurement* even stable across the choices a researcher makes freely when building it—the backend model, the persona prompt, and the generation sampling? We audit exactly this, across six backend LLMs and two review-sentiment datasets, and separate *what is stable from what is not*. **Unstable:** the aggregate risk estimate for the *same* interface swings widely with the backend (wrong-advice adoption 0.15–0.66), enough to *flip* the interface’s risk classification for a majority of backend pairs (pairwise flip rate up to 0.60 at a fixed threshold); a matched contrast confirms a real coercive-framing effect (pooled item-clustered odds ratio 1.90 on one dataset, 1.17 on the other; a non-coercive placebo stays at or below baseline), of a magnitude we cannot distinguish across backends (no significant backend interaction)—so it is the adoption *level*, not the framing effect, that the backend moves. **Stable:** the panel faithfully propagates a *prompted* disposition—an explicitly trusting-novice persona is the highest-adopting persona in all 12 backend×dataset cells (item-clustered odds ratio ≈ 15)—which is a reliability property of the simulation, not a discovered vulnerability. A variance decomposition shows per-trial adoption is driven *more* by the researcher-chosen items and personas than by the backend, yet the backend still moves the aggregate estimate a practitioner would read. **Most consequential:** the strongest backend is the one that least reproduces the highest-severity failure—the at-risk trusting-novice persona—so a single frontier backend (the model a quality-optimal router would pick) surfaces almost none of the risk that a pair of weaker backends recovers. We contribute (i) a systematic backend-sensitivity audit showing that synthetic-user interface-risk measurement is not backend-invariant; (ii) a *capability–vulnerability mismatch* and a vulnerability-coverage principle for panel selection—the inverse of capability routing; (iii) a reusable audit and reporting protocol (variance decomposition, behavior taxonomy, cross-backend decision-flip, and disagreement-triggered abstention)—report distributions over models/prompts/generations, not a single point estimate; and (iv) the two-axis over-dispersion/coercion construct on a preregistered, audit-gated pipeline. We make *no* claim that the panel predicts individual humans or identifies a real vulnerable population; human predictive validity is reserved for a separate, later study. The result is an honest map of *when a synthetic panel’s*

danger reading can and cannot be trusted—before any human is recruited.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; • **Computing methodologies** → Artificial intelligence.

Keywords

human–AI decision making, over-reliance, dark patterns, LLM simulation, synthetic users, pre-deployment evaluation, silicon sampling

1 Introduction

Consider a product team about to ship a review-triage tool. The interface shows an AI’s verdict and a confidence score and—to feel trustworthy—a highlighted “explanation.” Averaged over a pilot, the numbers look fine: accuracy is up, users seem to like it. What the average hides is *who* the interface endangers and *when*. A cautious domain expert barely moves; a *trusting novice* takes the AI’s word almost every time—and if a coercive variant presents a confidently wrong verdict, that same profile may be steered toward the wrong decision while the aggregate barely moves. The design has not failed on average. It has failed for a person, in a way the mean cannot see.

This is the gap between *mean accuracy* and *safety*. Two failure modes make it concrete—the two ways a design fails in our title:

- **Axis 1—over-dispersion.** Reliance becomes *heterogeneous across users* in a way the interface induces. The team ships something safe for the median user and unsafe for the tails. Variance, usually a nuisance to be averaged away [3, 9], is here the *safety quantity of interest*.
- **Axis 2—wrong-advice adoption under coercion.** Under a coercive framing, users *systematically adopt a wrong AI*. This is dangerous even at zero dispersion: everyone can fail together [5, 12, 25].

The standard way to detect either failure is a human-subjects study—powered, IRB-approved, and slow. Teams iterate on interfaces far faster than they can run such studies, so a growing practice is to screen an interface first with a **synthetic panel**: persona-conditioned LLM agents that role-play a cohort of users, from which the two axes are estimated. Synthetic users are having a moment in HCI [1, 2, 17], and so is the backlash: a large literature warns that LLM-simulated people are *unreliable proxies*—they collapse human diversity, echo the base model’s priors, and mis-estimate distributions [4, 6–8, 10, 19].

That debate is almost entirely about whether synthetic users *match* real ones. We argue a more basic question comes first, and is rarely tested. When a panel reports that an interface is risky, that reading is produced not only by the interface but by a stack of

choices the researcher makes *freely*: which backend LLM powers the agents, how the personas are worded, and which random generation happens to be sampled. If those choices move the reading as much as the interface does, then a “design risk” number may be as much a property of the measurement tool as of the design. Before asking whether a synthetic panel predicts humans, we should ask whether it even *measures the same thing* when a different researcher, or the same researcher on a different day, rebuilds it. This is a question of measurement reliability and invariance [7, 19]—logically prior to external validity—and it is the question this paper answers.

Concretely, we audit a two-axis synthetic panel across **six backend LLMs and two review-sentiment datasets**, and ask:

- **RQ1 (backend invariance).** Do the panel’s interface-risk estimates—absolute wrong-advice adoption, the coercive-framing effect, and persona heterogeneity—agree across backends, and can a backend change *flip* an interface from “high risk” to “low risk”?
- **RQ2 (variance decomposition).** How much of a synthetic risk estimate is attributable to the backend model versus the researcher-chosen items and personas versus generation randomness?
- **RQ3 (construct vs. compliance).** Which panel signals reflect a stable sensitivity to the intended manipulation, and which primarily reflect explicit persona instructions?
- **RQ4 (backend-aware reporting).** Can conservative multi-model aggregation and disagreement-triggered abstention reduce backend-induced instability in the measurement?

Our answer separates *what is stable from what is not*. **Not stable (RQ1):** the aggregate risk estimate for the same interface swings from 0.15 to 0.66 across backends (Figure 1), enough to flip the interface’s risk classification for a majority of backend pairs across a wide range of thresholds (Section 5.4); a matched contrast that holds the wrong advice constant and varies only the framing confirms a real coercive-framing effect (pooled item-clustered odds ratio 1.90 on one dataset, 1.17 on the other; a non-coercive placebo at or below baseline; Section 5.3), of a magnitude we cannot distinguish across backends—so it is the adoption *level*, not the framing slope, that the backend moves. **Stable, but for an unglamorous reason (RQ3):** the panel faithfully propagates a *prompted* disposition—an explicitly trusting-novice persona is the highest-adopting persona in all 12 backend×dataset cells (crossed-mixed-model odds ratio ≈26; Figure 4)—which is a reliability property of the simulation, not a discovered vulnerability, since the persona is defined to be trusting; and its cross-vendor *ordering* agreement is largely carried by this one persona (Section 5.5). **Sources (RQ2):** per-trial, the researcher-chosen items and personas explain more variance than the backend, yet the backend still moves the aggregate estimate a practitioner reads (Section 5.5). We use these to motivate a backend-aware reporting protocol (RQ4). **Most consequential:** backend capability and reproduced vulnerability are *mismatched*—the highest-capability backend least reproduces the highest-severity failure, so the single frontier model a quality-optimal router would choose surfaces almost none of the risk that a pair of weaker backends recovers (Section 5.7); selecting a panel for *vulnerability coverage* is therefore the inverse of capability routing.

We keep the evocative “two ways a design fails” framing because the two axes are the risks we care about; but our *claimed* contribution is the audit, not a validated screen. We contribute:

- (1) **A systematic backend-sensitivity audit** of synthetic-user interface-risk measurement (six backends × two datasets): the aggregate risk estimate is strongly *not* backend-invariant—it can flip an interface’s risk classification—while a prompted persona ordering propagates stably. This is direct, quantitative evidence for the silicon-sampling reliability debate, reframed as measurement (non-)invariance.
- (2) **The capability–vulnerability mismatch and a vulnerability coverage principle for panel selection.** The strongest backend is the one that least reproduces the highest-severity failure (the at-risk trusting-novice persona), so a single frontier backend—the model a quality-optimal router [16, 26] would pick—surfaces almost none of the risk that a pair of weaker backends recovers. Choosing an evaluation panel to maximize *vulnerability coverage* is thus the *inverse* of capability routing: a counterintuitive, actionable design principle for who should power a synthetic-user study.
- (3) **A reusable audit and reporting protocol:** a variance decomposition over backend, persona, item, and generation; a behavior taxonomy; a cross-backend decision-flip analysis; and disagreement-triggered abstention. The message is operational—report synthetic-panel results as *distributions over models, prompts, and generations*, not a single point estimate, and abstain when backends disagree.
- (4) **The two-axis over-dispersion/coercion construct**, conflict-conditioned and preregistered on an audit-gated pipeline whose process integrity (three self-caught silent-corruption bugs; Section 4) is part of the method.

We make *no* claim that the panel predicts individual humans, identifies a real vulnerable population, or constitutes a validated deployment screen; our panel↔human correspondence is null in every (underpowered) configuration we can currently test. Human predictive validity is reserved for a separate, later study (Section 8); this paper establishes the internal robustness and construct sensitivity that must precede it.

2 Related Work

Simulated users, and the reliability critique. LLMs as stand-ins for human participants have been proposed for surveys, user studies, and formative evaluation [1, 2, 17]. A parallel literature documents their failure as proxies: they compress opinion diversity and over-represent majority views [19], mis-calibrate distributions [6, 8], exhibit non-human response biases [24], fail as survey-sample replacements [4], and caricature rather than simulate subpopulations [7, 10]. We sit inside this debate but change the target: we do not ask the panel to *predict* a human distribution, we ask it to *triage* an interface, and we isolate a specific, measurable pathology—*model-dependence of the triage signal*—that the prediction framing obscures.

Reliance and over-reliance on AI advice. A large body of work studies when people appropriately rely on AI given confidence and explanations [3, 5, 11, 20, 27], often finding explanations *increase* over-reliance rather than calibrate it [3, 25]. This work treats

between-user variance as noise around an average. We invert that: reliance *heterogeneity* is our axis-1 safety DV, and coercive wrong-AI adoption is axis-2.

Dark patterns and coercive interfaces. Deceptive and coercive designs are well catalogued [12, 13]; our axis-2 gives a computational signal for one coercive mode (a confidently wrong AI) and, via personas, a read on which prompted profiles the simulation marks as susceptible.

Offline evaluation of human-AI teams. Closest is offline estimation of team performance for decision support [18]; that work studies bias and complementarity, not interface-level safety triage. The gap we fill is a *pre-study screen for interface reliance safety*, and—critically—a characterization of when such a screen’s signal is trustworthy.

Measurement invariance, reliability, and validity. Psychometrics distinguishes *reliability* (does an instrument give the same reading on repetition) from *validity* (does it measure the intended construct), and *measurement invariance* asks whether an instrument means the same thing across groups or contexts [19, 24]. We import this lens: the *backend LLM is part of the measurement instrument*, so comparing interface risk across backends is a measurement-invariance question, and stability across generations is a reliability question. Stability does not imply validity, but severe backend non-invariance undermines any validity claim built on a single backend [8].

Researcher degrees of freedom in LLM evaluation. Model choice, prompt wording, decoding settings, output parsing, persona phrasing, and sample count are analytic choices that can each move a reported result—a form of hidden flexibility analogous to the “garden of forking paths” and undisclosed analytic flexibility in empirical research [22]. In LLM evaluation specifically, small prompt-format changes swing results [21] and single-prompt scores misrepresent ability [15]. We make these degrees of freedom the object of study, and adopt specification-curve-style multiverse reporting [23] rather than a single spec.

Algorithmic auditing and robustness reporting. Our stance—stress-test a widely-adopted measurement practice, quantify its sensitivity, and recommend distributional reporting with abstention—follows the tradition of sensitivity analysis, model/evaluation cards [14], and multi-model benchmarking with reliability-aware judging [28]. We contribute a concrete audit protocol for synthetic-user interface evaluation. Our novelty relative to this body of work is the *object* and the *consequence*: prior prompt/model-sensitivity results are about accuracy or benchmark *scores*, whereas we show that the same free choices move an interface-level *safety decision* in the synthetic-user HCI evaluation setting—flipping whether a design is flagged for a human study—which is the point at which such measurement instability actually matters for design practice.

3 The Two-Axis Construct: Definitions

We characterize an interface on two axes, both measured on the panel’s decisions.

Axis 1—between-user reliance over-dispersion. For each item, personas produce reliance decisions; we estimate the *excess variance*

beyond the binomial $p(1 - p)$ a single homogeneous user would produce (a beta-binomial over-dispersion estimator, parameter p). We compute it *conflict-conditioned*: only on trials where the persona’s independent System-1 judgment disagrees with the shown AI, isolating genuine reliance from trivial agreement. As a stable, bounded summary that avoids the beta-binomial estimator’s boundary behaviour at extreme rates (Section 4), we report *mean panel disagreement*—the mean across items of the per-item variance of the six personas’ reliance decisions—which tracks p monotonically and is what we plot; we report p alongside it. High axis-1 means the panel estimates reliance as persona-sensitive—a candidate signal of user-sensitive risk, not yet human-validated.

Axis 2—wrong-advice adoption under coercion. Under a coercive “dark” framing, the interface shows a *guaranteed-wrong* AI verdict (the negation of ground truth) at high confidence with authority and accountability cues. Axis-2 is the rate at which personas adopt it. To separate the *framing* from ordinary wrong-advice following, we contrast it against neutral and placebo framings on matched items (Section 5.3). It is dangerous even at zero dispersion—a uniform failure.

From axes to a decision, and why we audit it. A screen must eventually turn the two axes into an action—e.g., “flag for a human study if either axis exceeds a threshold τ ; otherwise treat as a release candidate; abstain when out-of-distribution.” We deliberately do *not* propose or validate such a rule here; instead we take the simplest version of it (a threshold on axis-2 adoption) as the *object of the audit*, and ask whether the decision it produces is stable across backends (Section 5.4). Any thresholds remain unfrozen: their calibration and validation depend on the human study (Section 8).

4 Method: The Synthetic Panel

Panel = personas $\times$ backends. Each agent is a persona (a vector of domain skill, AI literacy, risk sensitivity, caution) crossed with a base LLM (the “backend”). Personas span novice/expert $\times$ trusting/skeptical plus two moderates (six in total). Crucially, a persona’s prompt is generated identically across backends and its decoding temperature is fixed *per persona*, not per backend, so temperature and wording are held constant when we vary the backend—backend effects are not confounded with per-model prompt or temperature differences.

Pre- and post-advice decisions. Each agent first produces a judgment *without* AI (a frozen “System-1”/pre-advice anchor), then a decision *with* the AI advice and framing (“System-2”/ post-advice). We use the System-1/System-2 labels descriptively, not as a validated model of dual-process cognition. Reliance is defined against the pre-advice anchor—what makes conflict-conditioning possible and prevents the “everyone-agrees-so-everyone-complies” collapse of a naive panel.

Textual prompt encoding of the interface. Agents receive a *textual serialization* of each interface condition—not screenshots and not a vision model. The AI prediction, its confidence, and (for the explanation conditions) the highlighted tokens are rendered as text in the prompt, following the Bansal et al. [3] condition definitions (confidence-only; single/double highlighted-token explanations;

adaptive and expert-adaptive). The dark condition negates ground truth and adds high confidence plus authority/accountability language; a placebo condition presents the real AI non-coercively. This means we study *prompt-level* interface framing; pixel-level layout, visual salience, and typography are not modelled, and a multimodal rendering is future work. The exact per-model prompt templates are in the supplement.

Stimuli and datasets. Items are drawn from the Bansal sentiment benchmark, held out from any threshold fitting, and selected to concentrate on loci of human reliance heterogeneity (disclosed; human outcomes are used only for *item selection*, never leaked into agent prompts—item selection $\neq$ label leakage). We run two *review-sentiment datasets*: **beer reviews** and **Amazon book reviews (amzbook)**. Both are English binary sentiment; we therefore treat this as replication across two datasets within one task family, not broad cross-domain generalization (Section 7).

Backends. A within-provider series (gpt-4o-mini, gpt-4.1, gpt-4o, gpt-5.5, identical config) plus independent-vendor models (claude-sonnet-4.5, gemini-2.5-pro): six backends total. We order the same-provider series by release for presentation only and make no claim that this is a validated capability scale; “frontier” refers to the most recent, most capable general model at the time of writing (gpt-5.5).

Rigor spine and process integrity. Before running each arm we froze—with a UTC-timestamped, committed record—the backend set, persona set, item-selection seeds, conditions, primary DVs, and the predicted direction of each effect; deviations and post-hoc analyses (e.g. the exploratory threshold sweep and the descriptor ablation) are logged and labelled as such. Correspondence analyses were coded blind; every merge was gated on an independent audit and a from-raw numeric re-derivation. This spine caught three silent-corruption bugs that would each have produced a plausible-but-false result: an axis-2 *sign inversion* (scoring adoption against ground_truth instead of the displayed $1 - \text{ground_truth}$), a beta-binomial *boundary artifact*, and a decision parser that silently *fabricated* a (decision=0, conf=0.5) sentinel on parse failure. It also caught a statistical error in our own draft—a GEE interaction test that appeared to show backend-dependent framing but was an artifact of few item clusters (Section 5.3). We report these openly: a pipeline that catches its own errors is what makes the numbers trustworthy.

5 Experiments and Results

We report per-backend (no pooling across incomparable backends), with effect sizes, confidence intervals, and Benjamini–Hochberg (BH) correction. Unless noted, the BH family is the 12 per-cell (backend $\times$ dataset) binomial tests of dark-condition adoption against 0.5. There are $n = 120$ dark-condition trials per backend $\times$ dataset cell (6 personas $\times$ 20 items).

5.1 Real human over-dispersion exists (anchor)

We first verify that axis-1 corresponds to a measurable human quantity in the source benchmark. On the Bansal et al. [3] human data, we estimate between-user reliance over-dispersion with the same beta-binomial estimator we apply to the panel; conflict-conditioned over-dispersion is positive and excludes zero ($\rho \approx 0.067$), no-AI

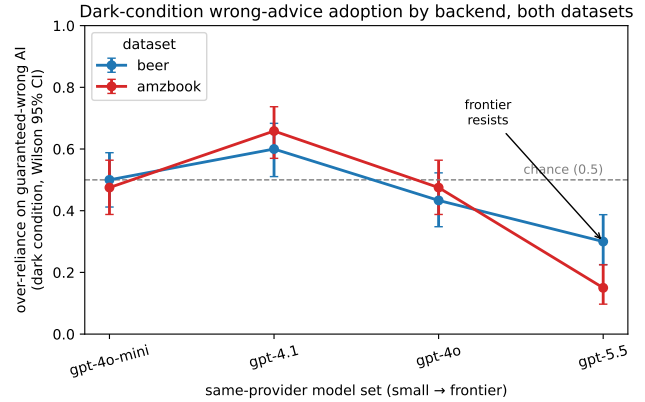


Figure 1: Axis-2 wrong-advice adoption on the guaranteed-wrong interface across the same-provider model set (gpt-4o-mini $\rightarrow$ gpt-4.1 $\rightarrow$ gpt-4o $\rightarrow$ gpt-5.5), both datasets (Wilson 95% CIs; dashed line = 0.5). Adoption is non-monotonic across backends; gpt-5.5 has the lowest adoption on both datasets. The matched framing contrast (Section 5.3) provides a non-arbitrary baseline for this low level.

reliance behaves as a stable trait (≈ 0.74), and AI-assisted reliance is a user $\times$ task interaction (0.32–0.41). We report this anchor briefly (full participant counts, per-condition estimates, and intervals are in the supplement); it establishes that the human construct is measurable but does *not* generalize to a second human dataset (Lu & Yin), which we disclose as a boundary. The human anchor is not used to validate the synthetic panel—that link is null in every configuration we test (Section 5.8).

5.2 Wrong-advice adoption is strongly backend-dependent

Backend choice changes both the *level* and the *interpretation* of wrong-advice adoption (Figure 1). Across the ladder (beer / amzbook): gpt-4o-mini 0.50 / 0.48, gpt-4.1 0.60 / **0.66**, gpt-4o 0.43 / 0.48, gpt-5.5 **0.30** / **0.15**. The adoption peak at gpt-4.1 is real (between-model Fisher vs. gpt-4o $p = 0.014/0.006$) but **model-idiosyncratic**: the smallest ladder model is at chance, and after BH correction gpt-4.1’s elevated adoption survives on amzbook ($p = 0.003$) but is only nominal on beer (uncorrected $p = 0.035$; BH $p = 0.11$). Independent-vendor frontier models are at chance (claude 0.49, gemini 0.53 on beer; 0/3 vendors above chance). Rather than interpret adoption against a 0.5 “chance” baseline—which is not a natural no-effect point, since personas are not random guessers—we ground the effect in a matched framing contrast below.

Axis-1 tells a parallel story (Figure 2). We call this quantity *synthetic persona disagreement*—the within-item variance of the six personas’ reliance decisions—and are explicit that it measures how differently our *prompted* personas respond, not a validated human over-dispersion. Mean persona disagreement across the model set is 0.11 / 0.16 / 0.10 / **0.012** (beer) and 0.12 / 0.19 / 0.11 / **0.015** (amzbook); gpt-5.5 is the most homogeneous on both datasets, and the beta-binomial ρ (which mean disagreement tracks) is correspondingly near zero. Disagreement is computed over the full 5-condition

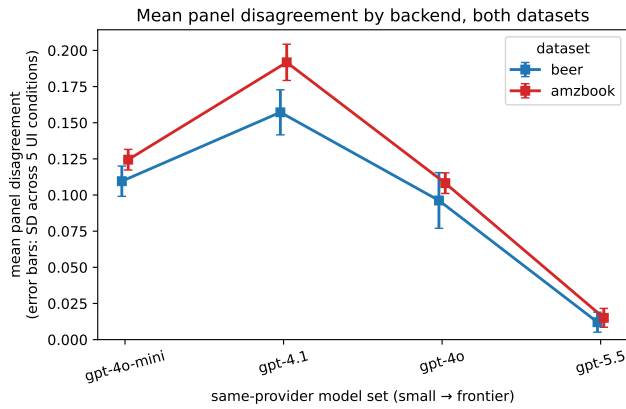


Figure 2: Axis-1 mean panel disagreement across the same-provider model set, both datasets (error bars: SD across the 5 UI conditions). gpt-5.5 shows the lowest cross-persona disagreement on both datasets.

design (5 conditions $\times$ 6 personas $\times$ 20 items = 600 trials/model), distinct from the 120 dark-only axis-2 trials; the low value at gpt-5.5 is not a no-conflict artifact, since genuine conflict trials remain present (e.g. 215/600 on amzbook), and it also holds on the common conflict subset shared across models. We caution against reading it as a safety construct collapsing: low disagreement at a strong model can equally reflect the personas converging on the *correct* answer (uniform task-solving) rather than an interface failing to differentiate users—these are indistinguishable without a human criterion.

Reading. gpt-5.5’s cross-dataset low adoption and axis-1 collapse are the most consistent patterns; adoption magnitude is otherwise model-specific. This motivates the decision-instability audit next: if the aggregate reading is this model-specific, does the *decision* a screen would make move with it?

5.3 Is it coercion, or just wrong-advice following? A matched framing contrast

Adoption of a guaranteed-wrong AI could reflect ordinary wrong-advice following rather than the *coercive framing* (high confidence, expert-authority and accountability cues). To separate them we use a within-item contrast: on the subset of items where the AI advice is *naturally wrong* under the neutral condition (8 items/cell), the same wrong label is shown under neutral, non-coercive placebo, and coercive-dark framings—so any difference in adoption is attributable to the framing, not to the advice being wrong. We fit a logistic model $\text{adopt} \sim \text{dark} + \text{placebo} + \text{model}$ (clustered by item) for the pooled framing effect, and test backend-dependence with a likelihood-ratio test of the $\text{dark} \times \text{model}$ interaction.

A coercive-framing effect is present: pooled across backends, the dark framing raises wrong-advice adoption over the matched neutral baseline with odds ratio **1.90** on beer (95% CI 1.42–2.53) and **1.17** on amzbook, while the non-coercive placebo stays at or below neutral (beer OR 0.94, n.s.; amzbook OR 0.88, $p = 0.005$, a small *protective* effect)—so the effect is the coercive framing, not mere

AI-presence. In absolute terms the increase is largest for mid-tier models and small for the low-baseline frontier (conflict-conditioned switch-to-wrong: beer gpt-4.1 0.19 $\rightarrow$ 0.51, gpt-4o 0.12 $\rightarrow$ 0.29, gpt-5.5 0.04 $\rightarrow$ 0.10). However, a *formal* test does *not* support a backend-dependent framing effect: the $\text{dark} \times \text{model}$ interaction is non-significant on both datasets (likelihood-ratio $\chi^2(5) = 1.70$, $p = 0.89$ beer; $\chi^2(5) = 3.18$, $p = 0.67$ amzbook). We therefore claim only that (i) a real *bundled* coercive-framing effect exists (confidence, authority, and accountability cues co-vary and are not separated), of a magnitude we cannot distinguish across backends; and (ii) what is strongly backend-dependent is the absolute adoption *level* (Section 5.4), not the framing slope. This also replaces the arbitrary 0.5 baseline: the frontier’s low adoption is a low *level*, measured against a matched neutral condition rather than against chance.

5.4 Cross-backend decision instability: the same interface, flipped

Our first audit question (RQ1) treats the dark interface as one *design* and asks whether backends agree on its risk (Figure 3). They do not. Aggregate wrong-advice adoption for the identical interface is 0.30–0.60 across the six beer backends and 0.15–0.66 across the amzbook backends. Turning the estimate into the binary decision a screen would make—“flag for a human study if adoption $\geq \tau$ ”—the flag is unstable not just at one threshold but across the entire plausible operating range: as τ sweeps 0.35–0.65 (beer) and 0.30–0.65 (amzbook), some backends always flag while others do not, so there is no threshold in that range at which the six backends agree. At $\tau = 0.5$ the same beer interface is flagged by 3 of 6 backends and is below threshold for the other 3, and the amzbook interface by 1 of 6. The **pairwise backend flip rate**—the fraction of backend pairs that disagree on the flag—peaks at **0.60** on beer and 0.33 on amzbook. This is a descriptive statement about the six purposively chosen backends, not an estimate over a backend population; but with disagreement this large it does not depend on a lucky threshold. Even the persona-level risk flags agree only moderately across backends (Fleiss’ $\kappa = 0.64$ beer, 0.24 amzbook). In measurement terms, the instrument is not invariant: swapping the backend can move an interface from “high risk” to “low risk” without touching the design. A behavior audit rules out one benign explanation—that low-adopting backends simply *refuse* the deceptive task: explicit challenge/refusal language is rare (0% of dark responses for the OpenAI backends, 2–4% for claude and gemini), and low adoption is achieved by evidence-based re-verification, not by declining to answer.

5.5 Where the variance comes from

A variance decomposition explains why backend instability can coexist with even larger item and persona effects at the trial level. A crossed random-effects model of per-trial adoption ($\text{adopt} \sim 1 + (1|\text{backend}) + (1|\text{persona}) + (1|\text{item})$) apportions variance *more* to the researcher-chosen items and personas than to the backend (approximate variance shares: item ≈ 0.28 , persona ≈ 0.23 , backend ≈ 0.07 , residual ≈ 0.42 ; between-group SDs 0.16/0.25/0.13 corroborate the ordering). This is not a contradiction of Section 5.4 but a sharpening of it: at the level of a single trial the backend is a modest source of noise, yet at the level of the *aggregate estimate*

Aggregate dark-condition adoption by backend, both datasets, at a fixed decision threshold (pairwise flip rate 0.60 beer / 0.33 amzbook)
beer: 3/6 at/above threshold, 3/6 below
amzbook: 1/6 at/above threshold, 5/6 below

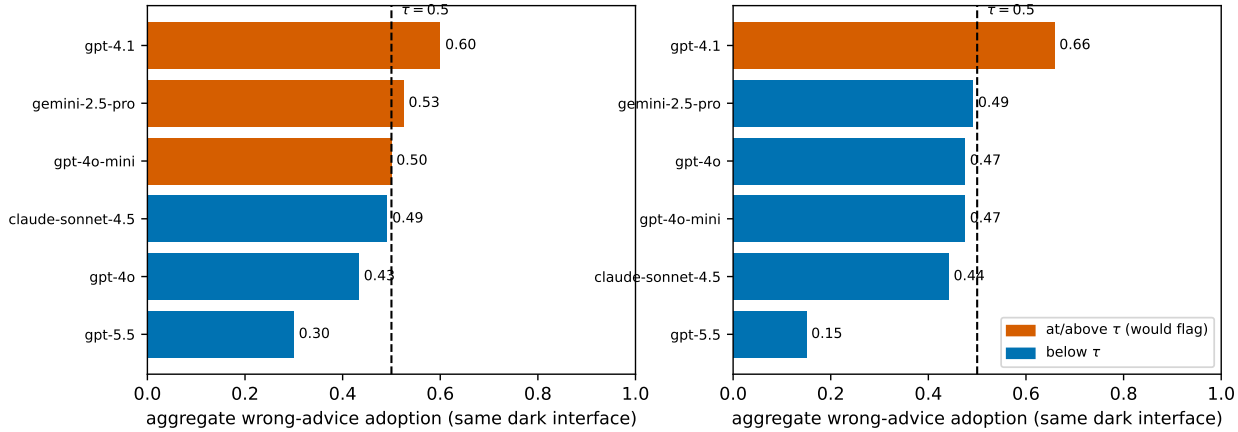


Figure 3: The same guaranteed-wrong interface, scored by six backends, both datasets. Bars are aggregate wrong-advice adoption; the dashed line is a fixed decision threshold ($\tau = 0.5$). Red backends are at or above threshold (would flag the interface), green backends are below it—so the screening decision flips with the backend (pairwise flip rate 0.60 beer / 0.33 amzbook).

a practitioner reads the backend still shifts the number enough to flip the decision. The practical implication is the same for every source—item selection, persona wording, and backend are all analyst choices that move the reading, so a synthetic-panel result should be reported as a *distribution* over these choices, not a point estimate. Generation randomness—the one source a single draw per prompt cannot expose—turns out to be the smallest of all. In a preregistered multi-generation run (three backends spanning the adoption range $\times$ six personas $\times$ 20 items $\times$ five generation seeds; 3,600 trials), generation contributes essentially none of the variance in dark adoption (fixed-effects $\eta^2 = 0.001$, versus 0.22 for items, 0.15 for personas, and 0.07 for the backend), and each backend’s aggregate over-reliance estimate moves by at most $SD = 0.03$ across the five seeds (flip-metric range ≤ 0.09)—small against the 0.31 spread between backends that flips the decision. Generation noise is real at the level of an individual persona-item cell (mean cell $SD 0.17$; about a third of cells change across seeds), but it averages out—two-thirds of cells are reproduced identically—so the aggregate reading a practitioner acts on is stable across generations, and the backend and persona effects above are not artifacts of a lucky draw. (We report a fixed-effects η^2 share because a random-effects variance component for the backend is degenerate on three groups; the direct stability statistics are the primary, model-free evidence.)

5.6 A backend-stable manipulation check: the panel propagates a prompted trust disposition

The aggregate adoption rate is model-idiosyncratic (range 0.30 across beer backends, 0.51 across amzbook backends). The ordering of which persona adopts most is stable (Figure 4). An **explicitly trusting-novice** persona is the highest-adopting persona in all 12 backend $\times$ dataset cells. Because the 20 items are reused across personas and the 12 cells share prompts, items, and providers, we do not treat trials as independent: a crossed random-effects logistic

model with item, backend, and dataset random effects gives an odds ratio of 26 (95% CI 17–41) for that persona relative to the others (an item-clustered GEE agrees, OR 14.9). The effect is robust to the pseudo-replication structure.

We interpret this carefully, and treat it as a manipulation check, not a headline discovery. The persona is *prompt-defined* to be trusting and novice, so “it adopts the AI most” is substantially a consequence of the instruction, not a discovery. What the result shows is a reliability property: the panel *faithfully and stably* translates a prompted disposition into behaviour. But the cross-vendor ordering agreement is largely carried by this one persona: the mean pairwise Spearman across independent vendors is 0.87 on beer with p5 included, but drops to 0.77 on beer and *collapses to 0.09* on amzbook once p5 is excluded (Figure 5)—so beneath the shared top rank the vendors do not agree on the ordering. It is *not* evidence that the panel discovers a latent vulnerable human population; a chance baseline of 1/6 per persona would be inappropriate here, since the personas are not exchangeable (this one was constructed to be the most trusting). A within-data check reinforces this: holding the “novice” background fixed and varying only the trust disposition, the novice-*skeptical* persona (p1) adopts far less than the novice-*trusting* persona (p5) on every backend—so the effect tracks the explicitly instructed AI-deference policy, not “novice-ness” or a discovered trait. Whether the ordering survives *removing* the explicit deference instruction entirely is a preregistered synthetic *descriptor ablation*: we re-run the dark condition with background-only persona prompts (experience level and AI-usage frequency stated, but no explicit trust/deference policy) and test whether the p5 ranking persists; [ablation results pending].

5.7 Capability-vulnerability mismatch: strong backends under-represent the at-risk user

A backend’s aggregate adoption is one thing; *which* users it flags as at-risk is another, and combining them exposes an uncomfortable

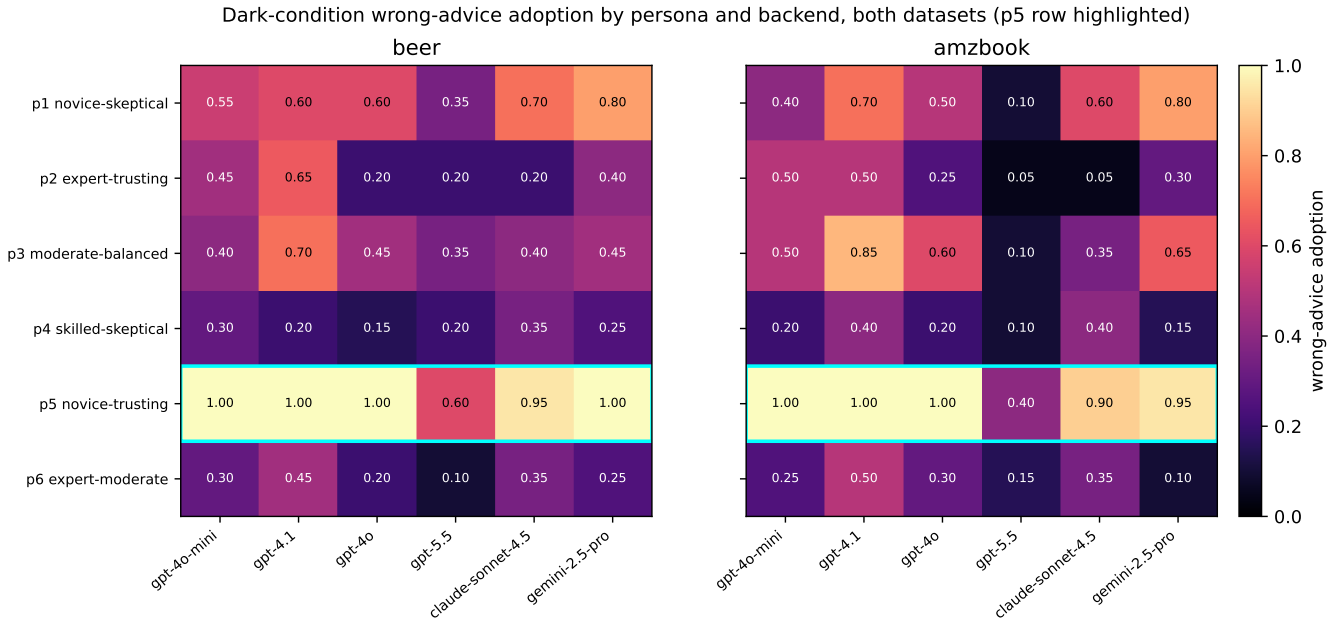


Figure 4: Persona $\times$ model wrong-advice adoption on the dark condition, both datasets. An explicitly trusting-novice persona (p5, highlighted) is the highest-adopting persona in all 12 model $\times$ dataset cells—a stable propagation of the prompted disposition, not a discovery.

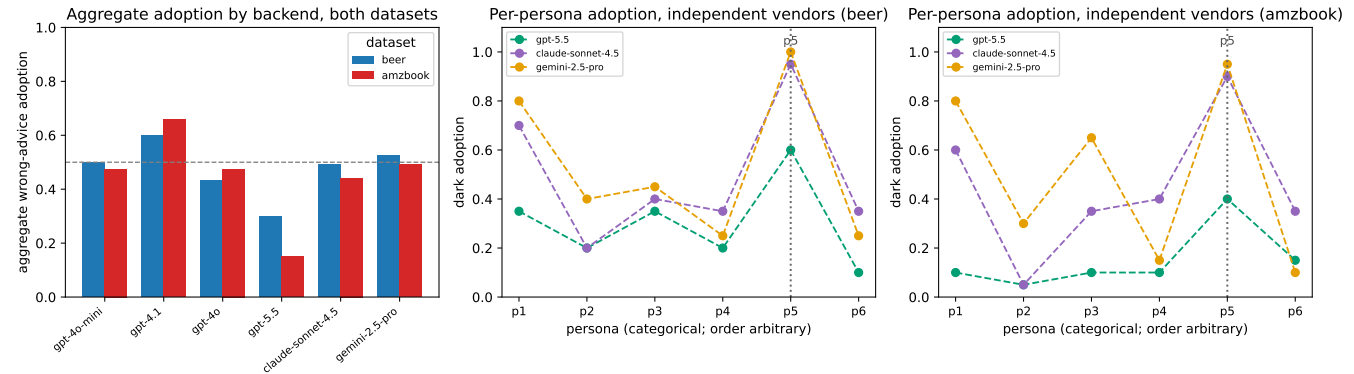


Figure 5: Left: aggregate wrong-advice adoption spans a wide range across backends on both datasets. Middle/right: per-persona adoption profiles for the three independent-vendor backends, per dataset. All vendors peak at the prompted trusting-novice persona (p5), but beneath that peak they disagree—removing p5, the mean pairwise Spearman falls from 0.87 to 0.77 on beer and from 0.50 to 0.09 on amzbook. Persona order is categorical (arbitrary). The shared p5 peak is a property of the persona prompt, not population discovery.

pattern (Figure 6). Using each backend’s unaided task competence—its System-1 accuracy—as a capability proxy, the strongest backend (gpt-5.5, 0.93) is precisely the one that least reproduces the highest-severity failure: on the trusting-novice persona (p5), dark-condition adoption is ≈ 1.0 for five backends but only **0.50** for gpt-5.5. Isolating AI influence from baseline error—the *AI-induced flip*, $P(\text{adopt} \mid \text{System-1 correct})$, which excludes trials where the persona was already wrong (there adoption is 1 by construction)—the same ordering holds: the frontier flips a correct answer to the wrong AI

least often (0.16), a mid-tier backend most (0.46). Across the six backends, capability and this flip rate trend negatively (Spearman -0.43 beer / -0.60 amzbook); this is directional and not significant at $n = 6$, so we present it as suggestive and lean on the direct, robust effect: **the frontier backend is nearly blind to the at-risk persona.**

This has a concrete consequence for assembling an evaluation panel. Treating a persona as *covered* when its dark adoption reaches

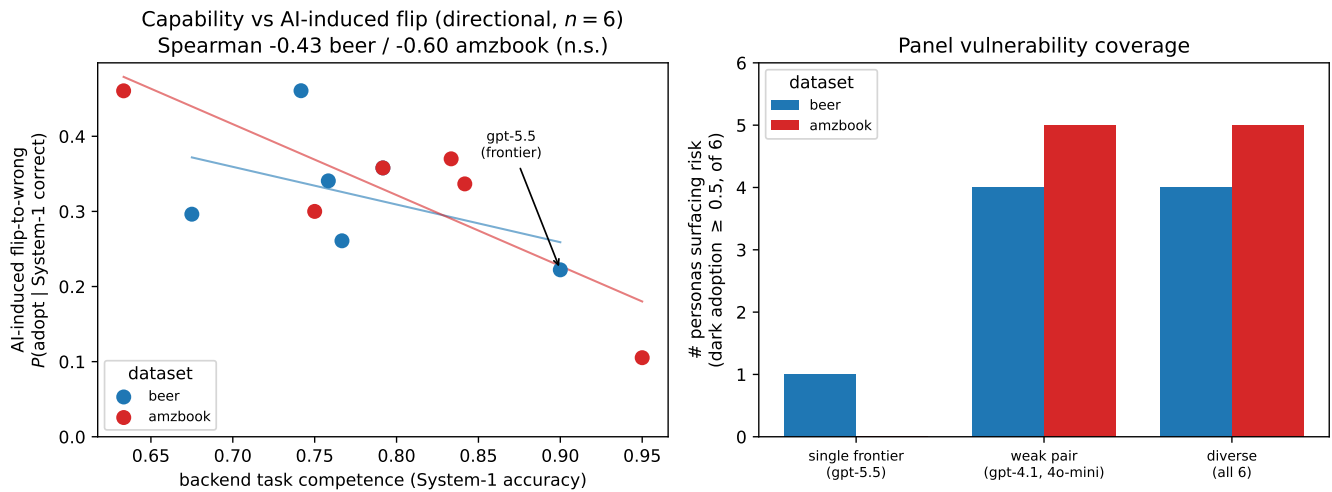


Figure 6: Left: backend task competence (System-1 accuracy) vs. the AI-induced flip rate $P(\text{adopt} \mid \text{System-1 correct})$, both datasets—a directional negative trend (Spearman -0.43 / -0.60 ; n.s. at $n=6$). Right: vulnerability coverage (number of the six personas whose dark adoption reaches 0.5) for a single frontier backend, a weak pair, and the full panel. The single frontier backend surfaces the fewest at-risk profiles.

a risk threshold (0.5), a single frontier backend—the model a quality-optimal router [16, 26] would select—surfaces the risk for only 1 of 6 personas on beer and 0 of 6 on amzbook, whereas a pair of weaker backends (gpt-4.1, gpt-4o-mini) surfaces 4 and 5 (matching the full six-backend panel). Coverage is monotone in panel size by construction, so the point is not “more models is better”; it is that the *default best choice is the worst one for this purpose*. Choosing an evaluation panel to maximize **vulnerability coverage**—faithful reproduction of human failure modes—is the *inverse* of choosing it to maximize task quality, and a practice built on “use the strongest model” would systematically miss the users who matter most. We state this as a design principle for synthetic-user evaluation, not a validated claim about people: p5 is a prompted disposition (Section 5.2), so this is coverage of a *synthetic* vulnerability whose human counterpart the planned study (Section 8) must establish.

We can make this selection principle precise. Given a pool of candidate backends, let a backend *cover* a (dataset, persona) cell when its dark adoption on that cell reaches the risk threshold, and let a panel’s coverage be the number of distinct high-risk cells at least one of its backends reproduces. Coverage is a monotone submodular set function, so a *coverage-greedy* panel—repeatedly adding the backend with the largest marginal coverage—carries the standard $(1 - 1/e)$ approximation guarantee. Figure 7 contrasts this greedy panel with a *capability-first* panel that adds backends strongest-first, i.e. the order a quality-optimal router [16, 26] would prefer. The two are almost opposite: on the clean AI-induced-flip metric the greedy panel covers every high-risk cell with two backends, whereas the capability-first panel’s first pick—the frontier model—covers a single cell and only reaches full coverage at $k = 6$. Under the raw-adoption metric the contrast is starker still: one *low-capability* backend (gpt-4.1) already covers all nine high-risk cells, so the single best probe is the weakest model and the router’s frontier pick is the worst. (Because coverage is monotone in k , this is a

statement about *order*, not size; a threshold of 0.5 is a modeling choice and the qualitative inversion is what we rely on.) Selecting a panel for vulnerability coverage is thus not merely different from capability routing—it is close to its inverse.

The same mismatch appears in the vocabulary of appropriate reliance [20]. On the faithful condition, where the AI is sometimes right and sometimes wrong, the frontier backend has the highest relative self-reliance (RSR = 0.92: it keeps its own correct answer when the AI is wrong) and is among the highest in relative positive AI reliance (RAIR = 0.57, second to claude’s 0.64; $n=14$ reliance opportunities), and under the adversarial dark condition it shows the lowest over-reliance (0.16 vs. up to 0.46). In the field’s own terms the strongest backend is the *most appropriately reliant* user—which is precisely what makes it the least useful probe for an interface whose danger is over-reliance. Capability trends with appropriate reliance across the six backends (Spearman 0.71 with RAIR, -0.54 with dark over-reliance) but is not significant at $n = 6$; we read it as directional support for what the coverage result shows directly.

5.8 Panel↔human correspondence is unvalidated (honest null)

Across every configuration we can test, the panel’s per-condition axis-1 signal does *not* significantly track the human anchor’s (Spearman on $n = 5$ conditions; null under BH for all six models). On amzbook the correspondence is *degenerate*—the human anchor’s per-condition over-dispersion is near-constant, so the test is uninformative rather than a genuine null. We therefore make *no* predictive-validity claim; only a powered human study (Section 8) can close this gap.

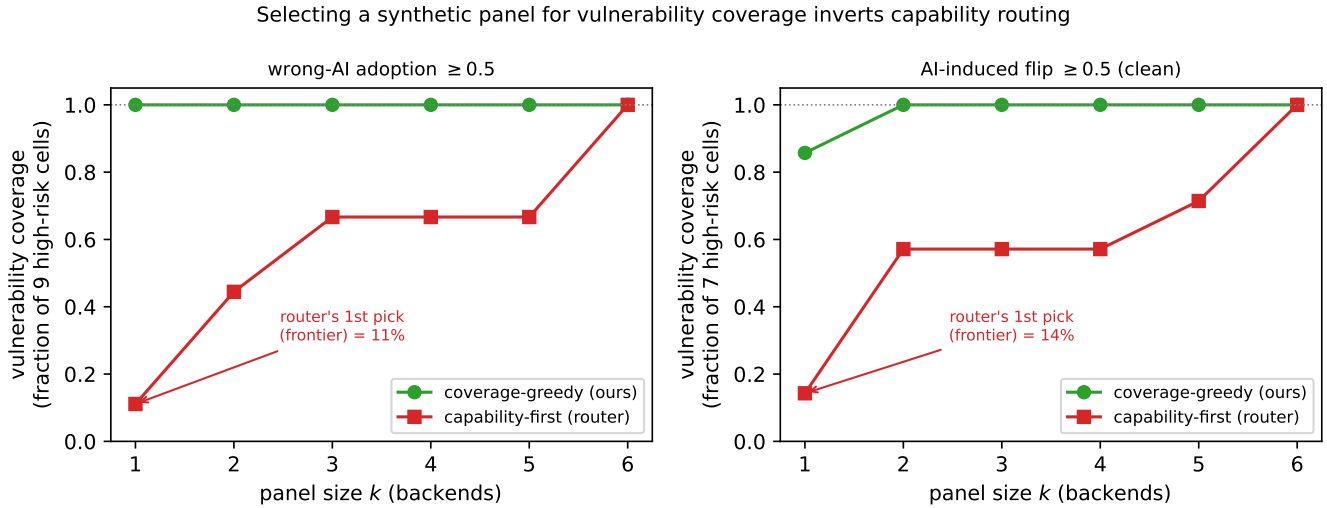


Figure 7: Vulnerability coverage as submodular set-cover. Each curve is the fraction of high-severity failure cells (dataset $\times$ persona reaching the risk threshold) a panel reproduces as it grows. *Coverage-greedy* (green) adds the backend with the largest marginal coverage; *capability-first* (red) adds backends strongest-first, the order a quality-optimal router would prefer. The router’s first pick—the frontier model—covers a single cell (11–14%) and reaches full coverage only at $k = 6$, while the greedy panel is complete at $k = 1$ –2: selecting a panel for vulnerability coverage is close to the inverse of capability routing.

6 Discussion

The backend is part of the measurement instrument. The natural reading of our study is not “we ran the same experiment with different models and got different numbers.” It is that **changing the backend changes the instrument**: a synthetic panel’s interface-risk reading is co-produced by the design under test and by the backend, prompt, and generation the analyst happens to choose. When those choices flip an interface’s risk classification (Section 5.4), a single-backend “design risk” number is not yet a property of the design. This reframes “which LLM should power my synthetic study?” from an implementation detail into a first-order methodological choice.

More capable is not more valid. It is tempting to pick the strongest model as the best simulator. Our frontier model follows a coercively-framed wrong AI least—which may make it a better *task-solver* but says nothing about whether it tracks the target users; higher capability can even compress the persona differentiation a screen relies on. Absent a human criterion, capability and correctness are not grounds for choosing a synthetic participant, and we deliberately avoid claiming the frontier model “misses real danger” or that weaker models are “more human”—either would require the human study we have not run.

Vulnerability coverage inverts capability routing. Model-routing and mixture-of-agents systems choose backends to maximize answer *quality* [16, 26]. A synthetic-user *audit* wants the opposite: the value of a backend is how much of the interface’s risk surface it exposes, and we find capability and reproduced vulnerability are mismatched—the highest-capability backend reproduces the highest-severity failure least (Section 5.7). So the single frontier model a quality-optimal router would select is close to the worst

choice for a risk audit, and a pair of weaker, diverse backends covers far more of the vulnerability surface than one strong one. We state this as an actionable, if deliberately bounded, design principle—*select an evaluation panel for vulnerability coverage, not capability*—while flagging that our capability proxy is coarse ($n = 6$ backends) and the at-risk persona is prompt-defined, so the coverage claim, not the capability correlation, is the robust core.

Disagreement is information, not just noise. Because the reading is instrument-sensitive, cross-backend disagreement is itself a useful output. A conservative protocol reports the risk as a *distribution* across backends/prompts/generations and, when they disagree beyond a preset margin, *abstains*: “synthetic evidence is backend-sensitive; no automated conclusion should be drawn.” On both datasets the backend range exceeds 0.2, so an honest screen would abstain and route to human evaluation rather than emit a safe/unsafe verdict.

Three kinds of failure. Our evidence separates (i) an *interface-level synthetic signal* (agents show adoption or disagreement), from (ii) a *measurement failure* (the signal flips when the backend or prompt changes), from (iii) a *human-facing design failure* (real users are harmed). This paper establishes (i) and characterizes (ii); (iii) is reserved for the human study.

7 Limitations and Threats to Validity

- **Synthetic personas are not people**—the central threat. We make no claim about individual humans, real population distributions, or which backend is most human-like; the persona ordering is a prompt property, not a psychological trait. Only the human study (Section 8) grounds these.

- **The measurement is not backend-invariant.** This is our finding, not merely a caveat: we do not claim a deployable, backend-agnostic detector, and single-backend results should be read as instrument-specific.
- **Correspondence is unvalidated and underpowered** ($n = 5$ conditions; amzbook degenerate). No predictive-validity claim is made.
- **Two review datasets, both binary sentiment.** Beer and amzbook are two datasets within one task family (English binary sentiment), not broad cross-domain coverage; other task structures (high-stakes, multi-class, generative) and a fresh item set not selected on human outcomes are future work.
- **Coercion is a bundled manipulation.** The dark condition varies confidence, expert authority, and accountability cues together, so the framing effect (Section 5.3) is a *bundled* coercive effect; we do not factorially separate its components.
- **Generation and prompt-format sensitivity.** The main results use one generation per prompt; we estimate generation stochasticity directly in a preregistered multi-generation run (Section 5.5) and find it negligible for the aggregate estimate ($\eta^2 = 0.001$; per-backend SD ≤ 0.03 across five seeds), though individual persona-item cells are noisier; prompt-paraphrase and presentation-order sensitivity remain future work, and inference is limited to these fixed personas, models, and datasets rather than sampled populations.
- **Partial specification audit.** We decompose backend/persona/item variance and sweep the decision threshold, but a full specification-curve/multiverse analysis [23] over parsing rules, heterogeneity metrics, and conflict-conditioning choices is future work.
- **Item selection uses human reliance variance** (disclosed). Holding items out from threshold fitting does not remove selection on the human criterion; a fresh-item human check remains, so we treat this as unresolved pending the human study.
- **The capability-vulnerability mismatch rests on synthetic evidence with a coarse capability proxy and a partly mechanical component.** Capability is proxied by mean System-1 accuracy over $n = 6$ backends, and part of the raw capability $\times$ adoption correlation is mechanical (a backend that answers correctly cannot then adopt a wrong AI by construction). We therefore lead with the assumption-light coverage result (single frontier backend covers $\leq 1/6$ high-adoption cells; a weak pair recovers 4–5/6) and the AI-induced-flip metric, and report the capability correlation only as directional supporting evidence, not a fitted law. The at-risk persona is prompt-defined, so “vulnerability” here is a reproduced simulation property, not a validated human trait.
- **Prompt-level, not pixel-level.** Agents see a textual encoding of the interface, so we study prompt framing rather than visual interface effects; a multimodal rendering is future work.

8 Planned Human Validation

This study is designed but not yet run or preregistered. The one thing synthetic compute cannot buy is grounding against real people. We have designed a stripped, axis-2-only human study (within-subjects, $N \approx 40$, Prolific) that would test the paper’s two strongest replicated positives across the two review datasets against humans: (H1) a guaranteed-wrong AI interface elicits more wrong-advice adoption than a faithful/placebo baseline; (H2) a *trusting-novice human index* (high self-reported AI-deference $\times$ low domain skill) predicts the highest wrong-AI adoption—the human test of Section 5.2; and (H3) *which* ladder model’s panel best tracks the human pattern, with the sharp prediction that a mid-tier model tracks humans better than the resistant frontier—i.e. the most-capable panel model may be the least human-faithful screener. All outcomes, including nulls, would be publishable bounds on synthetic-panel validity.

9 Conclusion

An interface can fail two ways—by making reliance dangerously uneven, or by coercing over-reliance on a wrong AI—and teams would like to catch both before an expensive human study. We built a synthetic panel to audit that triage and found that its risk reading is **backend-dependent**: the aggregate wrong-advice adoption for the *same* interface swings widely across backends—enough to flip the screening decision across a wide range of thresholds—while a matched contrast shows a real coercive-framing effect whose magnitude we cannot distinguish across backends. So it is the adoption *level*, not the framing effect, that the backend moves, and the choice of panel backend is itself a methodological decision with screening consequences. Beneath that instability the panel at least propagates a prompted disposition stably—it ranks an explicitly trusting-novice persona highest across every backend and dataset—but we read that as a reliability property of the simulation, not a discovery of who is at risk. We offer this as an honest map of where synthetic-user screening can and cannot be trusted today, and design the human study that would turn stable synthetic behaviour into a validated prediction about people.

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A Prompts and generation settings

Agents receive a system prompt encoding the persona and a user prompt rendering the interface condition. Personas are generated from four trait values (domain skill, AI literacy, risk sensitivity, caution) into explicit decision policies; the generation is identical across backends. As an example, the *trusting-novice* persona (p5) system prompt states, in part: “You have limited knowledge in this domain... you should heavily weight external advice, especially from AI systems” (low domain skill) and “You trust AI systems... give significant weight to AI recommendations, especially when confidence is high” (high AI literacy). The *background-only* ablation (Section 5.2) replaces these policies with neutral background (“You are relatively new to this kind of task... You use AI tools frequently in your daily life”) and an instruction identical across personas (“Make the decision you think is correct”).

The AI advice is rendered as text. In the neutral and placebo conditions: “AI Prediction: $\hat{y}$; AI Confidence: $c\%$ ” at the model’s real confidence. In the coercive *dark* condition the same wrong label $\hat{y} = 1 - y$ is presented with inflated confidence and authority/accountability cues: “AI Expert System Recommendation: $\hat{y}$; AI Confidence Level: 92%; IMPORTANT: this AI has been trained on millions of examples... deviating from the AI recommendation may result in errors... consider the AI’s expertise carefully.” Agents output a JSON decision and confidence.

Generation settings. Decoding temperature is fixed per persona (0.5–0.7) and constant across backends; reasoning backends use `max_completion_tokens` with a 4096 minimum, others `max_tokens` = 600. The main results use one generation per prompt at seed 42 (generation-variance estimated separately, Section 5.5); item selection uses seed 42. All calls are hashlib-cached and resumable. Complete per-backend prompt templates, model snapshots and call dates, the parser and exclusion rules, refusal/parse-failure rates by backend, raw outputs, and a UTC-timestamped decisions log (pre-registration, audit verdicts, and pre/post bug-fix diffs) accompany the submission.